

ANSWER KEY — Integration Lesson 32 Test: Partial Fractions

For the grader · every split verified by re-adding the pieces · every integral verified by differentiating · credit correct method with small arithmetic slips

1. Write as a single logarithm: (a) $\ln|x - 5| - \ln|x + 2|$ (b) $\ln 12 - \ln 4$ (c) $3 \ln 2$ (d) $2 \ln|x - 1| + \ln|x + 4|$

Work: the three rules: $\ln a + \ln b = \ln(ab)$, $\ln a - \ln b = \ln(a/b)$, $k \cdot \ln a = \ln(a^k)$.

(a) subtraction $\rightarrow$ division: $\ln|(x - 5)/(x + 2)|$.

(b) $\ln 12 - \ln 4 = \ln(12/4) = \ln 3$. (Numeric check: $2.485 - 1.386 = 1.099 = \ln 3$ ✓.)

(c) coefficient $\rightarrow$ exponent: $3 \ln 2 = \ln(2^3) = \ln 8$. (Check: $3 \times 0.693 = 2.079 = \ln 8$ ✓.)

(d) first the coefficient: $2 \ln|x - 1| = \ln((x - 1)^2)$; then addition $\rightarrow$ multiplication: $\ln((x - 1)^2|x + 4|)$. (Since $(x - 1)^2$ is never negative, the bars on it are unnecessary.)

Answer: (a) $\ln|(x - 5)/(x + 2)|$ (b) $\ln 3$ (c) $\ln 8$ (d) $\ln((x - 1)^2|x + 4|)$.

Grader note: classic errors — writing (b) as $\ln 8$ (subtracting the insides instead of dividing) and (c) as $\ln 6$ (multiplying the inside by 3 instead of raising it to the 3rd power). In (d), accept $\ln|(x - 1)^2(x + 4)|$ as well.

2. Find A and B: $7/((x - 3)(x + 4)) = A/(x - 3) + B/(x + 4)$, then verify by re-adding.

Work: multiply both sides by $(x - 3)(x + 4)$: $7 = A(x + 4) + B(x - 3)$, true for EVERY x.

$x = 3$ (kills the B term, since $3 - 3 = 0$): $7 = A(3 + 4) = 7A \rightarrow A = 1$.

$x = -4$ (kills the A term): $7 = B(-4 - 3) = -7B \rightarrow B = -1$.

Verify by re-adding: $1/(x - 3) - 1/(x + 4) = [(x + 4) - (x - 3)] / ((x - 3)(x + 4)) = (x + 4 - x + 3) / ((x - 3)(x + 4)) = 7/((x - 3)(x + 4))$ ✓.

Answer: $A = 1, B = -1 \rightarrow 7/((x - 3)(x + 4)) = 1/(x - 3) - 1/(x + 4)$.

Grader note: the standard error is a sign-flipped magic value — using $x = -3$ for the factor $(x - 3)$, or $x = +4$ for $(x + 4)$. Guard: substitute the magic value into its factor and confirm it gives 0. Full credit requires the re-adding check to be shown, since the test asked for it.

3. Find A and B: $3x/((x - 4)(x + 2)) = A/(x - 4) + B/(x + 2)$, and verify by re-adding.

Work: clear denominators: $3x = A(x + 2) + B(x - 4)$.

$x = 4$: left side $3(4) = 12$; right side $A(4 + 2) = 6A \rightarrow 12 = 6A \rightarrow A = 2$.

$x = -2$: left side $3(-2) = -6$; right side $B(-2 - 4) = -6B \rightarrow -6 = -6B \rightarrow B = 1$.

Verify by re-adding: $2/(x - 4) + 1/(x + 2) = [2(x + 2) + 1(x - 4)] / ((x - 4)(x + 2)) = (2x + 4 + x - 4) / ((x - 4)(x + 2)) = 3x/((x - 4)(x + 2))$ ✓.

Answer: $A = 2, B = 1 \rightarrow 3x/((x - 4)(x + 2)) = 2/(x - 4) + 1/(x + 2)$.

Grader note: the new trap versus question 2 is that the top must also be evaluated at the magic value — forgetting to plug x into the $3x$ (e.g. writing $3 = 6A \rightarrow A = 1/2$) is the expected slip. Also watch the double negative at $x = -2$: $-6 = -6B$ gives $B = +1$, not -1 .

4. Find $\int \frac{4}{(x-1)(x+3)} dx$.

Work: f'/f check first: the bottom expands to $x^2 + 2x - 3$ with derivative $2x + 2$, but the top is 4 — no match, so partial fractions it is.

Split: $4 = A(x+3) + B(x-1)$. $x = 1$: $4 = 4A \rightarrow A = 1$. $x = -3$: $4 = B(-4) \rightarrow B = -1$.

Verify the split: $1/(x-1) - 1/(x+3) = [(x+3) - (x-1)]/((x-1)(x+3)) = 4/((x-1)(x+3))$ ✓.

Integrate each piece ($\int 1/(x-a) dx = \ln|x-a| + C$): $\int = \ln|x-1| - \ln|x+3| + C = \ln|(x-1)/(x+3)| + C$.

Check by differentiating: $d/dx[\ln|x-1| - \ln|x+3|] = 1/(x-1) - 1/(x+3) = 4/((x-1)(x+3))$ ✓ — the integrand.

Answer: $\ln|x-1| - \ln|x+3| + C$ ($= \ln|(x-1)/(x+3)| + C$).

Grader note: both the split form and the combined single-log form earn full credit. Deduct for missing absolute-value bars only as a note, not a full point, at this stage. A common wrong answer is $4 \ln|x-1| - 4 \ln|x+3|$ (the 4 was never absorbed into A and B).

5. Find $\int \frac{2}{x(x+2)} dx$.

Work: a bare x is a linear factor too, so its piece is A/x and its magic value is $x = 0$.

Template: $2/(x(x+2)) = A/x + B/(x+2) \rightarrow$ clear: $2 = A(x+2) + Bx$.

$x = 0$: $2 = A(2) \rightarrow A = 1$. $x = -2$: $2 = B(-2) \rightarrow B = -1$.

Verify the split: $1/x - 1/(x+2) = [(x+2) - x]/(x(x+2)) = 2/(x(x+2))$ ✓.

Integrate: $\int = \ln|x| - \ln|x+2| + C = \ln|x/(x+2)| + C$.

Check by differentiating: $1/x - 1/(x+2) = 2/(x(x+2))$ ✓.

Answer: $\ln|x| - \ln|x+2| + C$ ($= \ln|x/(x+2)| + C$).

Grader note: the point being tested is that the magic value for the factor x is 0 (not 2, and not "there isn't one"). Watch for students who write the template as $A/x + B/x$ — one piece per *distinct* factor, and the second factor is $(x+2)$.

6. Find $\int (2x + 3)/(x^2 + 3x - 10) dx$; name the tool and say why partial fractions is the slow road.

Work: run the decision habit — differentiate the bottom first: $d/dx[x^2 + 3x - 10] = 2x + 3$, which is exactly the top. So this is the f'/f pattern from Lesson 25, and $\int f'/f dx = \ln|f| + C$ finishes it in one step.

$$\int (2x + 3)/(x^2 + 3x - 10) dx = \ln|x^2 + 3x - 10| + C.$$

Check by differentiating: $d/dx[\ln|x^2 + 3x - 10|] = (2x + 3)/(x^2 + 3x - 10)$ ✓ (chain rule: $1/\text{inside} \times \text{derivative of inside}$).

Why partial fractions is the slow road: the bottom does factor, as $(x + 5)(x - 2)$, so splitting is *legal* — it just costs three extra steps (factor, solve for A and B, integrate two logs) to reach the same place. Doing it that way: $2x + 3 = A(x - 2) + B(x + 5)$; $x = -5$ gives $-7 = -7A \rightarrow A = 1$; $x = 2$ gives $7 = 7B \rightarrow B = 1$, so $\int = \ln|x + 5| + \ln|x - 2| + C = \ln|(x + 5)(x - 2)| + C = \ln|x^2 + 3x - 10| + C$ — the identical answer, via the log addition rule.

Answer: $\ln|x^2 + 3x - 10| + C$, using $\int f'/f = \ln|f|$ (the top is exactly the derivative of the bottom). Partial fractions gives $\ln|x + 5| + \ln|x - 2| + C$, the same thing, with triple the work.

Grader note: full credit for either route with a correct answer, but the question explicitly asks the student to *notice* the shortcut — award the point only if they identify f'/f and explain it. $\ln|x + 5| + \ln|x - 2| + C$ is a correct final answer. Watch for a spurious $1/2$ out front (there is no constant to fix here: the top matches f' exactly, not $2f'$).

7. Find $\int (3x + 1)/(x^2 - x - 6) dx$.

Work — f'/f check first: $d/dx[x^2 - x - 6] = 2x - 1$, and the top is $3x + 1$. Not a match, not even off by a constant factor $\rightarrow$ no shortcut. Proceed with partial fractions.

Factor the bottom: two numbers multiplying to -6 and adding to -1 are -3 and $+2$, so $x^2 - x - 6 = (x - 3)(x + 2)$. (Expand to check: $x^2 + 2x - 3x - 6 = x^2 - x - 6$ ✓.)

Template: $(3x + 1)/((x - 3)(x + 2)) = A/(x - 3) + B/(x + 2) \rightarrow 3x + 1 = A(x + 2) + B(x - 3)$.

$x = 3$: $3(3) + 1 = 10 = A(5) \rightarrow A = 2$. $x = -2$: $3(-2) + 1 = -5 = B(-5) \rightarrow B = 1$.

Verify the split: $2/(x - 3) + 1/(x + 2) = [2(x + 2) + (x - 3)]/((x - 3)(x + 2)) = (2x + 4 + x - 3)/(x^2 - x - 6) = (3x + 1)/(x^2 - x - 6)$ ✓.

Integrate: $\int = 2 \ln|x - 3| + \ln|x + 2| + C$. (Single-log form: $\ln((x - 3)^2|x + 2|) + C$.)

Check by differentiating: $d/dx[2 \ln|x - 3| + \ln|x + 2|] = 2/(x - 3) + 1/(x + 2) = (3x + 1)/(x^2 - x - 6)$ ✓.

Answer: $2 \ln|x - 3| + \ln|x + 2| + C$.

Grader note: expected errors — factoring as $(x + 3)(x - 2)$ (right product, wrong signs: that expands to $x^2 + x - 6$); and evaluating the top wrong at $x = -2$ ($3(-2) + 1 = -5$, not -7 or $+5$). Note the second log has NO coefficient — a student who writes $2 \ln|x - 3| + 2 \ln|x + 2|$ has copied A onto B.

8. Compute $\int_0^1 \frac{2}{(x+1)(x+3)} dx$ exactly, as a single log.

Work: decompose: $2 = A(x+3) + B(x+1)$. $x = -1$: $2 = A(2) \rightarrow A = 1$. $x = -3$: $2 = B(-2) \rightarrow B = -1$.

Verify the split: $1/(x+1) - 1/(x+3) = [(x+3) - (x+1)]/((x+1)(x+3)) = 2/((x+1)(x+3)) \checkmark$.

Antiderivative: $\ln(x+1) - \ln(x+3)$. On $[0, 1]$ both $x+1$ and $x+3$ are positive, so the absolute-value bars can be dropped. (The blow-up points $x = -1$ and $x = -3$ are safely outside $[0, 1]$.)

FTC — at $x = 1$: $\ln 2 - \ln 4$. At $x = 0$: $\ln 1 - \ln 3 = 0 - \ln 3 = -\ln 3$.

Integral = $(\ln 2 - \ln 4) - (-\ln 3) = \ln 2 - \ln 4 + \ln 3 = \ln(2 \cdot 3 / 4) = \ln(6/4) = \ln(3/2)$.

Sense-check: $\ln 1.5 \approx 0.405$. The integrand runs from $2/3 \approx 0.67$ down to $2/8 = 0.25$ over a width-1 interval, so a value around 0.4 is exactly right $\checkmark$.

Answer: $\ln(3/2) \approx 0.405$.

Grader note: accept $\ln 3 - \ln 2$, $\ln 1.5$, or $\ln(6/4)$ — all the same number. Classic errors: forgetting $\ln 1 = 0$ and writing the lower value as $-\ln 3 + 1$; and subtracting the endpoints backwards, which gives $\ln(2/3)$ (negative — impossible, since the integrand is positive on $[0, 1]$). A decimal-only answer misses the "exactly" instruction.

9. Find $\int \frac{5x}{(x-1)(x+4)} dx$, then write it as one single logarithm.

Work: f'/f check: bottom = $x^2 + 3x - 4$, derivative $2x + 3 \neq 5x \rightarrow$ no shortcut.

Template: $5x/((x-1)(x+4)) = A/(x-1) + B/(x+4) \rightarrow 5x = A(x+4) + B(x-1)$.

$x = 1$: $5(1) = 5 = A(5) \rightarrow \mathbf{A = 1}$. $x = -4$: $5(-4) = -20 = B(-5) \rightarrow \mathbf{B = 4}$.

Verify the split: $1/(x-1) + 4/(x+4) = [(x+4) + 4(x-1)]/((x-1)(x+4)) = (x+4+4x-4)/((x-1)(x+4)) = 5x/((x-1)(x+4)) \checkmark$.

Integrate: $\int = \ln|x-1| + 4 \ln|x+4| + C$.

Combine: $4 \ln|x+4| = \ln((x+4)^4)$ (a 4th power is never negative, so no bars needed), then add: $\ln(|x-1|(x+4)^4) + C$.

Check by differentiating: $d/dx[\ln|x-1| + 4 \ln|x+4|] = 1/(x-1) + 4/(x+4) = 5x/((x-1)(x+4)) \checkmark$.

Answer: $\ln|x-1| + 4 \ln|x+4| + C = \ln(|x-1|(x+4)^4) + C$.

Grader note: the sign trap is at $x = -4$: $-20 = -5B$ gives $B = +4$, not -4 . Also watch the log-power rule going the wrong way — $4 \ln|x+4|$ is $\ln((x+4)^4)$, not $\ln(4(x+4))$ or $4 + \ln|x+4|$. Accept $\ln|(x-1)(x+4)^4|$.

10. $f(x) = (x + 7)/(x^2 + 2x - 3)$: (a) factor and decompose; (b) $\int f \, dx$; (c) $\int_2^3 f \, dx$ exactly; (d) as a single log.

(a) f'/f check: derivative of $x^2 + 2x - 3$ is $2x + 2 \neq x + 7 \rightarrow$ partial fractions.

Factor: two numbers multiplying to -3 and adding to $+2$ are $+3$ and $-1 \rightarrow x^2 + 2x - 3 = (x - 1)(x + 3)$. (Expand: $x^2 + 3x - x - 3 = x^2 + 2x - 3$ ✓.)

Template: $(x + 7)/((x - 1)(x + 3)) = A/(x - 1) + B/(x + 3) \rightarrow x + 7 = A(x + 3) + B(x - 1)$.

$x = 1$: $1 + 7 = 8 = A(4) \rightarrow \mathbf{A = 2}$. $x = -3$: $-3 + 7 = 4 = B(-4) \rightarrow \mathbf{B = -1}$.

Verify by re-adding: $2/(x - 1) - 1/(x + 3) = [2(x + 3) - (x - 1)]/((x - 1)(x + 3)) = (2x + 6 - x + 1)/(x^2 + 2x - 3) = (x + 7)/(x^2 + 2x - 3)$ ✓.

(b) $\int f \, dx = 2 \ln|x - 1| - \ln|x + 3| + C$. **Check:** $d/dx = 2/(x - 1) - 1/(x + 3) = (x + 7)/(x^2 + 2x - 3)$ ✓.

(c) On $[2, 3]$ both $x - 1$ and $x + 3$ are positive (and the blow-ups at $x = 1$ and $x = -3$ are outside), so use $2 \ln(x - 1) - \ln(x + 3)$.

At $x = 3$: $2 \ln 2 - \ln 6$. At $x = 2$: $2 \ln 1 - \ln 5 = 0 - \ln 5 = -\ln 5$.

Integral $= (2 \ln 2 - \ln 6) - (-\ln 5) = 2 \ln 2 - \ln 6 + \ln 5$.

(d) $2 \ln 2 = \ln 4$, so the total is $\ln 4 + \ln 5 - \ln 6 = \ln(4 \cdot 5 / 6) = \ln(20/6) = \mathbf{\ln(10/3)} \approx 1.204$.

Sense-check: $f(2) = 9/5 = 1.8$ and $f(3) = 10/12 \approx 0.83$, so the average height over the width-1 interval is around 1.2 — matching $\ln(10/3) \approx 1.20$ ✓.

Answer: (a) $2/(x - 1) - 1/(x + 3)$ (b) $2 \ln|x - 1| - \ln|x + 3| + C$ (c) $2 \ln 2 - \ln 6 + \ln 5$ (d) $\ln(10/3) \approx 1.20$.

Grader note: award the parts separately. Expected errors: factoring as $(x + 1)(x - 3)$ (that expands to $x^2 - 2x - 3$); $B = +1$ instead of -1 ($4 = -4B$); dropping the $2 \ln 1 = 0$ term at the lower limit; and in (d) turning $2 \ln 2$ into $\ln 2^2 = \ln 4$ incorrectly as $\ln 8$ or $2 \cdot 2$. Accept any exact equivalent of $\ln(10/3)$, such as $\ln 20 - \ln 6$ or $\ln 10 - \ln 3$.